Final Year Project Short Proposal

Cloud Policy Cost and Security Simulator

Maryam Shahzad (2212396)
Muhammad Abdur Rehman Khan (2212399)
Wasfa Nauman Bhatti (2312462)

Supervised by: Mr. Hasnain Tahir

Department Computer Science
Faculty of Computing and Engineering Sciences
SZABIST University
Islamabad, Pakistan

Spring 2026

Introduction

The proposed system will be an intelligent platform designed to address the critical gaps in cloud governance, financial operations, and cybersecurity awareness for students and SMEs in Pakistan. As cloud adoption accelerates, local users struggle with the complexity of compliance, cost estimation, and security postures without access to practical, hands on tools. This proposal outlines the development of a Cloud Policy, Cost and Security Solution, a unified platform that provides a risk free simulation environment. Its overarching goal is to bridge the gap between theoretical knowledge and practical application by offering a "Digital Twin" of a cloud ecosystem.

Purpose

The purpose of the platform will be to revolutionize cloud education by integrating Artificial Intelligence for smarter, risk free operational training. It will aim to solve issues regarding the lack of practical exposure to governance, cost planning, and security protocols. The system will empower users to master cloud operations, predict anomalies, and enforce policies through intelligent agents, fulfilling the fundamental need for developing skills in AI driven security operations centers (SOCs).

Targeted Audience

The primary audience will include university students and SMEs in Pakistan who seek a practical understanding of cloud infrastructure without facing high costs. It will also serve DevOps and AIOps professionals interested in the intersection of security and automation. These users will value a localized platform to experiment with governance rules, cost management strategies, and cybersecurity protocols in a controlled, sandboxed environment.

Product Functions and Features

The system will deliver a comprehensive simulation of cloud operations integrated with advanced Machine Learning algorithms. Features will include an Intelligent Governance Engine using Natural Language Processing to parse user policies and enforce standards automatically. The platform will offer a Predictive FinOps module employing time series forecasting to anticipate costs and optimize resources. Crucially, it will incorporate AI driven security using Deep Neural Networks (DNN) for threat detection and Reinforcement Learning agents for autonomous remediation, creating a self healing environment. It will also provide visualization tools to display risk scores and compliance reports.

Social and Industrial Impact

The platform will enhance digital literacy by significantly upgrading the technical skill set of Pakistani students in cloud security and AIOps. It will drive better employment outcomes by bridging the gap between theoretical knowledge and industry requirements. On an industrial level, it will enable SMEs to validate security strategies prior to deployment, reducing the likelihood of costly data breaches. The system will foster a culture of responsible cloud usage and demonstrate the potential of AI in solving complex operational challenges.